called **1-cocycles** (of A with coefficients in $\mathrm{Hom}_k(W,V)$). They form a vector space denoted by $Z^1(W,V)$.

(b) Let $X : W \to V$ be a linear map. The coboundary of X, dX, is defined to be the function $A \to \mathrm{Hom}_k(W,V)$ given by $dX(a) = \rho_V(a)X - X\rho_W(a)$. Show that dX is a cocycle which vanishes if and only if X is a homomorphism of representations. Thus coboundaries form a subspace $B^1(W,V) \subset Z^1(W,V)$, which is isomorphic to $\mathrm{Hom}_k(W,V)/\mathrm{Hom}_A(W,V)$. The quotient $Z^1(W,V)/B^1(W,V)$ is denoted by $\mathrm{Ext}^1(W,V)$.

(c) Show that if $f, f' \in Z^1(W,V)$ and $f - f' \in B^1(W,V)$, then the corresponding extensions U, U' are isomorphic representations of A. Conversely, if $\phi : U \to U'$ is an isomorphism such that

$$\phi = \begin{pmatrix} 1_V & * \\ 0 & 1_W \end{pmatrix},$$

then $f - f' \in B^1(V,W)$. Thus, the space $\mathrm{Ext}^1(W,V)$ "classifies" extensions of W by V.

(d) Assume that W, V are finite dimensional irreducible representations of A. For any $f \in \mathrm{Ext}^1(W,V)$, let U_f be the corresponding extension. Show that U_f is isomorphic to $U_{f'}$ as representations if and only if f and f' are proportional. Thus isomorphism classes (as representations) of nontrivial extensions of W by V (i.e., those not isomorphic to $W \oplus V$) are parametrized by the projective space $\mathbb{P}\mathrm{Ext}^1(W,V)$. In particular, every extension is trivial if and only if $\mathrm{Ext}^1(W,V) = 0$.

Problem 3.9.2. (a) Let $A = \mathbf{C}[x_1, \ldots, x_n]$, and let V_a, V_b be 1-dimensional representations in which the elements x_i act by a_i and b_i, respectively ($a_i, b_i \in \mathbf{C}$). Find $\mathrm{Ext}^1(V_a, V_b)$ and classify 2-dimensional representations of A.

(b) Let B be the algebra over $\mathbf{C}$ generated by $x_1, \ldots, x_n$ with the defining relations $x_i x_j = 0$ for all i, j. Show that for $n > 1$ the algebra B has infinitely many nonisomorphic indecomposable representations.

Problem 3.9.3. Let Q be a quiver without oriented cycles, and let P_Q be the path algebra of Q. Find irreducible representations of P_Q and compute Ext^1 between them. Classify 2-dimensional representations of P_Q.